

Comment

The missing link among the functions of the temporo-parietal junction

Comment on “Left and right temporal-parietal junctions (TPJs) as “match/mismatch” hedonic machines: A unifying account of TPJ function” by Doricchi et al.

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Correctly identifying the elements of the outside world that match or mismatch our internal representation of it is crucial in many aspects of our life, from daily activities requiring appropriate interpretation of a social situation to efficient selection of potentially relevant information for our goals.

In this review, Doricchi et al., [1] propose a unifying account for the role of one of the most puzzling areas in the human brain, the temporo-parietal junction (TPJ), as well as for the different findings around a well-established lateralization of its functions into the left and right hemispheres. Anatomically, the TPJ has been found to have heterogeneous histological and connectional profiles, while functionally it is characterized by ubiquitous involvement in high-level cognition ranging from social reasoning to memory retrieval and attentional orienting. Interestingly, some of these functions have been seen to be lost after selective lesion of the right/left, or related to activation limited to only its right/left side in functional-MRI (fMRI) studies. This high diversity of results has further increased the mystery around a possible common functional role of this area in cognition.

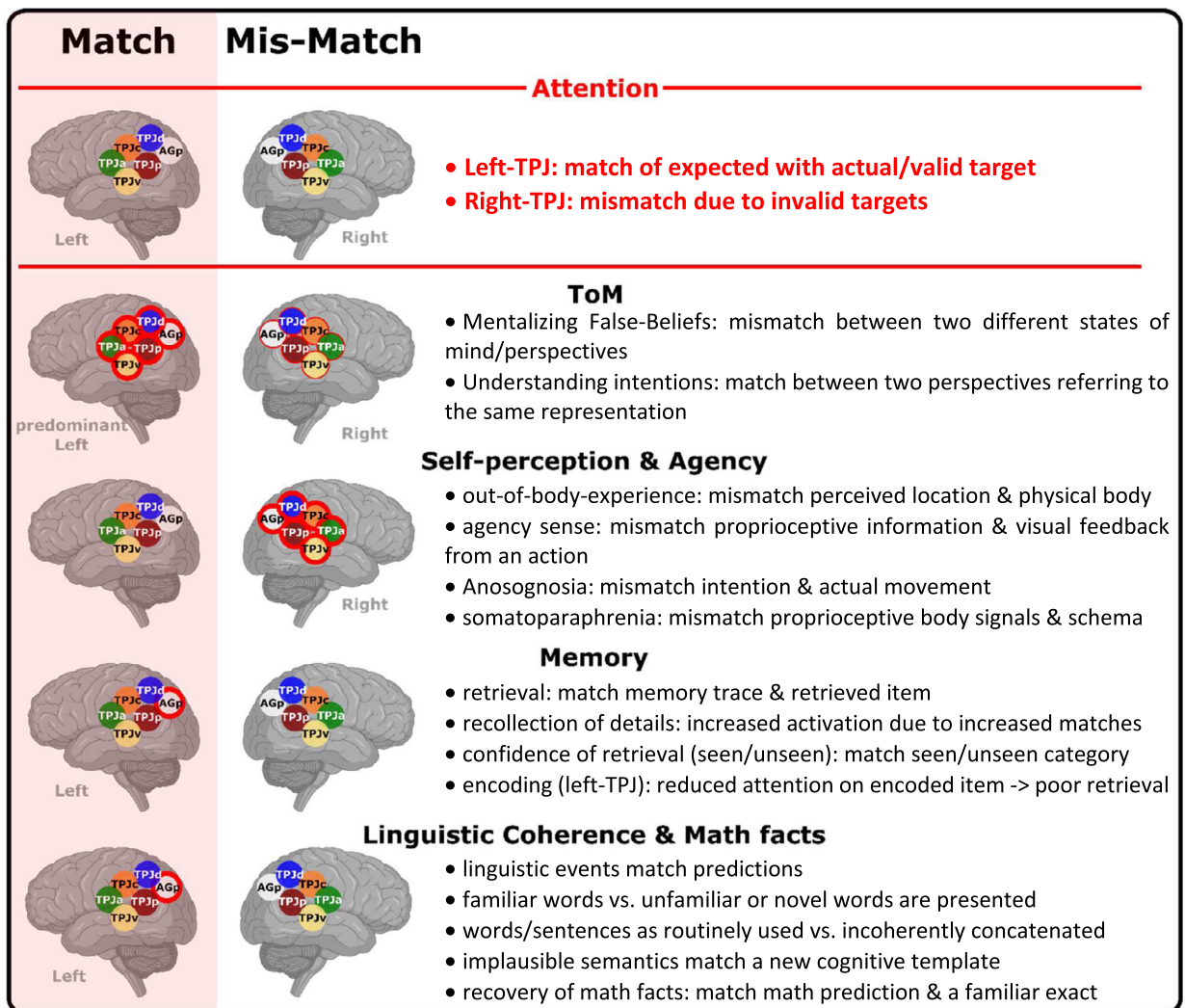
Stemming from an experiment that taps into the validity and invalidity of spatial attentional cuing [2] and from the recent independent replications of similar observations in the domain of feature-based attention [3,4], Doricchi proposes that the right TPJ codes mismatches between expected and actual target events, i.e., invalid targets, while the left TPJ codes both mismatches and matches, i.e., invalid and valid targets, respectively (Fig 1, top row).

Doricchi then reviews results on the ubiquitous involvement of TPJ in seemingly unrelated cognitive domains and seeks to explain divergent accounts on TPJ lateralization within his novel match-mismatch framework (Fig 1). Across the five domains of Attention, Theory of Mind (ToM), Self-perception & Agency, Memory, and Linguistics & Math coherence, activations of and lesions to left TPJ seem to always accord with an account in terms of matches between expected and actual events, while activations of and lesions to right TPJ seem to accord with mismatches between expected and actual events.

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As testified by the current issue, the classic attentional account of TPJ has played a critical role in explaining its role and fueling debate on a possible specialization across all of these domains [5]. This is perhaps because attention is a core cognitive function that contributes to many other domains of information processing or because it has a long history of investigations and thus holds the record of published studies on this region.

An interesting difference between TPJ investigations in the attention compared to the other domains is the strong connection to stimuli explicitly presented in at least one (sometimes more) sensory modalities that characterizes the literature on attention. Visual and to some extent auditory or even tactile stimuli have been extensively used in Posner-like experiments highlighting TPJ involvement in attention (e.g. [6–8]). These sensory stimuli are typically placed in an abstract space, with very limited links to real-life elements (such as objects) or contexts (such as realistic environments). On the contrary, research on ToM, self-perception, language, and to some degree memory, usually pivots on more abstract thinking (rather than perceiving) ability that nevertheless bears on realistic situations, body presence, semantic knowledge, or realistic memorized objects. This double-dichotomy between actual sensory stimuli vs. more abstract mental objects on the one hand, and between abstract space vs. realistic environmental space on the other, possibly makes TPJ even more fascinating.



The sensory-based knowledge characterizing attentional literature has a counterpart in the field of neuropsychology. Visuo-spatial tasks in brain-damaged patients have uncovered the existence of the spatial neglect syndrome, a puzzling neuropsychological condition wherein patients present an impaired capacity to perceive and respond to sensory stimuli

located in the contralesional space (i.e. left side after right hemisphere damage). This deficit arises during either spontaneous or cued reorienting of attention towards external stimuli, often results from lesion in (right) TPJ [9], and generally fits well the match-mismatch theory proposed by Doricchi et al., (see paragraph 5.5 and Appendix A). More interestingly, the use of more complex or naturalistic object-like stimuli in visual tests in patients has highlighted the existence of an object-centered (or allocentric) form of spatial neglect, wherein patients display impaired processing of the contralesional side of a given visual object (e.g. reporting only half of a shape or half of a word), independently of the absolute spatial position of these objects. Damage to TPJ (e.g. [9–11]), and perhaps more specifically to ventral temporal regions [12,13], has been associated with this object-centered component of neglect. Moreover, a recent fMRI investigation has proposed the existence of a ventro-temporal attention area [14], possibly overlapping with lesion sites leading to object-centered deficits in some neglect patients. Yet, because of the sparsity of object-based measures in attention literature and neuropsychology, a clear link between TPJ role in object-based attention cannot be definitely drawn (but see [15] for preliminary evidence), and even less so a link with the match-mismatch framework of TPJ as proposed by Doricchi.

Objects are sensory and physical stimuli with several conceptual and semantic properties attached to them. These properties make objects optimal candidates to investigate questions across diverse cognitive domains. As a matter of facts, objects are often time the protagonists of ToM, language, and memory experiment. Deeper investigations on object-based attention might thus help overcome some of the experimental differences that exist across the studies highlighting TPJ involvement in cognition. Moreover, objects come in many forms, colors or orientation and can therefore be precisely controlled, adapted and employed across different experimental frameworks. Additionally, because they are simple, ubiquitously present, intuitive entities, objects are also suitable for testing animal models – while it would be hard to investigate these models with very abstract approaches such as those employed in ToM or out-of-body experience (e.g. [16]). Objects might thus bear the potential of putting us a step forward into one other still unresolved gap around TPJ, the lack of a homologous area in animal models [17].

In lights of these considerations, it would be interesting to test whether the proposed match-mismatch framework still holds in studies where attention is allocated to more realistic items such as meaningful objects or to more realistic spatial context, but also test whether this object-based experimental frame can better reconcile TPJ functions across cognitive domains, whether other recently reported areas (e.g. [14]) might complement the current picture, and whether a match-mismatch area may be found and explored in animal models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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